BiG-FSLF: A Cross Heterogeneous Domain Few-Shot Learning Framework Based on Bidirectional Generation for Hyperspectral Image Change Detection

Xianghai Wang, Siyao Li, Xiaoyang Zhao, and Keyun Zhao

***Abstract*— In recent years, hyperspectral image change detection (HSI-CD) based on deep learning has achieved high detection accuracy, but these methods obtain excellent detection results usually rely on having sufficient labeled samples to train the network. However, the production of HSI labels is difficult, costly, and inefficient. In practical tasks, often only a limited number of labeled samples can be obtained due to the limitation of timeliness. To address this problem, a cross-heterogeneous domain few-shot learning framework based on bidirectional generation (BiG-FSLF) is proposed for HSI-CD, which aims to solve the few-shot problem of HSI-CD by few-shot learning (FSL) and to assist HSI-FSL to perform better by obtaining learnable changed information (i.e., empirical knowledge) from another remote sensing data. Specifically, a multitask generation encoder (MLGenE) is designed to take on both the tasks of FSL and domain adaptation to achieve HSI-CD under the condition of cross-heterogeneous domain few-shot. First, we take any pair of image data in a very high-resolution image (VHRI) CD dataset as the source domain and HSI is used as the target domain, using sufficient labeled samples in the source domain and a small number of labeled samples in the target domain for FSL. Meanwhile, a bidirectional generation domain adaptation (BiGDA) method based on a generative adversarial strategy is proposed to achieve adaptive alignment of the two heterogeneous domains (source and target domains) feature distributions, to mitigate the impact of the domain shift problem inherent to cross-domain data on FSL. Abundant experiments with only five training samples on the publicly available popular HSI-CD datasets confirm that the proposed method can show great detection performance. The source code of the proposed framework will be released at https://github.com/lsylnnu/BiG-FSLF.**

***Index Terms*— Change detection (CD), convolutional neural network (CNN), few-shot learning (FSL), generative adversarial, hyperspectral image (HSI).**

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I. Introduction

REMOTE sensing image change detection (CD) can provide timely surface-changed information, and it is an important technique for identifying surface differences by comparing a group of images taken multiple times in the same coverage area [1], [2]. Hyperspectral images (HSIs) have high spectral resolution, so such image data, which provides rich information about the spectral characteristics of the surface, can more easily capture finer ground object change than other images. Based on this characteristic, CD technology has been greatly promoted and developed [3]. At present, using HSI to extract changed information is becoming an inevitable choice in the CD field, and the HSI-CD technology has achieved a series of achievements in many fields, such as land resource survey, urban development planning, vegetation ecological monitoring, and environmental detection [4], [5], [6], [7], [8], [9], [10].

In recent years, the deep learning method has received widespread attention in various fields due to its significant advantages in deep feature representation and nonlinear modeling, and has shown great potential in the field of HSI-CD [11]. Among the various architectures of deep learning, the convolutional neural network (CNN) is one of the most typical ones, which builds a deep network model by convolutional layers and pooling layers in a layer-by-layer construction, and each layer can extract the image corresponding feature information. At the same time, it enables end-to-end detection of input images on the network, avoiding complex preprocessing of the image. For example, Zhan et al. [12] designed a CNN that could decompose the hyperspectral change tensors in three directions, which used convolutions of different dimensions to extract features of HSI from multiple directions. Ding and Li [13]. proposed a joint spatial–spectral–temporal attention method for HSI-CD, which could use a convolutional block to get spatial and spectral feature representation. An improved CNN network based on the pyramidal convolutional structure was proposed by Yang et al. [14] in order to enable the network to handle changed details of ground object features at different scales. Although the above methods all show some excellent performance in HSI-CD to varying degrees, their good performance often relies on training the network with a large number of training samples. However, collecting enough labeled training samples is impractical in most remote sensing tasks, especially for HSIs, land cover

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experts are usually required to conduct field surveys, which greatly increases the cost and time of sample annotation.

For solving the problem of limited labeled samples, unsupervised learning has attracted much attention because it does not require labeled data, and some unsupervised HSI-CD methods have emerged in recent years. For example, papers [15], [16], and [17] were dedicated to improving the sample generation strategy in order to generate high-quality pseudo-labels, and then using the generated pseudo-label training set to complete the training of the network, which provided new paradigms for the implementation of unsupervised HSI-CD. Although the unsupervised method does not require labeled information for feature learning, the generated pseudo-labels for training are usually different from the real labeled dataset. This difference will interfere with the actual training, making it difficult for the model to capture complex changes, which greatly affects the accuracy of the final detection result.

In fact, the information provided by the small number of labeled samples in the dataset can help improve the discrimination of the network and should be effectively utilized in the training process. Recently, few-shot learning (FSL) has gained much attention for its ability to help alleviate the burden of collecting large amounts of labeled data. FSL refers to the idea of learning from a small number of labeled samples so that the model can obtain the ability to solve tasks using very few labeled samples [18]. It has been used to solve the problem of limited labeled samples in the field of hyperspectral classification. For instance, Zhang et al. [19] proposed a global prototypical network (GPN). The network was trained by a global representation learning strategy, and then the unclassified samples could be classified by computing the similarity between them and each global prototypical representation. Tong et al. [20] proposed an attention-weighted graph convolutional network (AwGCN), where the attention-weighted graph was used to fully quantify the relationships of all samples, and subsequently, the graph convolutional network was used to extend the labels. Eventually, the class of sample was determined based on the trained attention-weighted graph. Li et al. [21] proposed an FSL framework named GDMFSL, which was a combination of graph-based and deep multitask networks (DMN). GDMFSL integrated graph information into the DMN to assist in learning the potential relationships between all samples alleviated the overfitting problem caused by limited labeled samples, and improved the quality of detection.

Currently, FSL-based methods are used for HSI classification tasks and have not been explored in the HSI-CD field. According to researchers [22], [23], the accumulation of empirical knowledge and the transformation of a large amount of relevant data knowledge are especially important for FSL. Therefore, it is a more recognized trend to assist few-shot data by introducing knowledge of the related fields. With the continuous development of remote sensing technology, although the labeled information of the dataset to be detected (called target domain) is scarce, other available CD datasets are increasing. So, it is now possible to use a remote sensing image with sufficient labeled data as a data source (called source domain) to provide transferable knowledge to assist the target domain for prediction [24], [25]. However, there is still a problem that because the source and target data are captured by different sensors and there are differences in land cover scenarios, such cross-domain data do not have the same intradomain distribution, called the domain shift problem, which will prevent effective transfer of learnable knowledge.

To address the above problems, a cross-heterogeneous domain FSL (CHFSL) framework based on bidirectional generation (BiG-FSLF) is proposed for HSI-CD. We use a pair of images in a very high-resolution image (VHRI)-CD dataset as the source domain, which requires simpler technology for obtaining data, lower cost, and easier to label (this VHRI is not from the same region as the HSI, so no registration is required). HSI is the target domain, so FSL has performed alternately on the cross-heterogeneous domain data to learn generalizable knowledge of the source domain data and personality features of the target domain data. At the same time, an effective solution is also proposed for the domain shift problem inherent in cross-heterogeneous domain data, that is, bidirectional generation domain adaptation (BiGDA) based on a generative adversarial strategy to achieve adaptive alignment of two domain feature distributions and finally achieve HSI-CD under the condition of cross heterogeneous domain few-shot. The main contributions can be grouped into the following points.

1) A CHFSL framework based on bidirectional generation is proposed for HSI-CD. As far as we know, FSL and generative adversarial strategy are unified into a CD framework for the first time, which is a pioneering research work to assist HSI to realize CD through VHRI data.
2) We design a multitask generation encoder (MLGenE), which can simultaneously undertake the two tasks of FSL and feature distribution reconstruction during domain adaptation, aiming to better complete FSL under the condition of the cross-heterogeneous domain.
3) A BiGDA method is proposed, which provides an effective solution to the domain shift problem of heterogeneous data.

The remaining sections of this article are scheduled as listed below. The related works about the application of FSL methods based on the meta-learning strategy for HSI processing and the methods for domain adaptation will be reported in Section II. A detailed description of the proposed method will be written in Section III. In Section IV, the experimental datasets, related settings, and experimental results are described and analyzed, and the corresponding ablation experiments are performed for the proposed framework. In Section V, we will discuss the visualization results of the proposed method, the number of training samples, and the choice of hyperparameters. Section VI is the conclusion.

II. Related Work

A. Meta-Learning Strategy in FSL for HSI Processing

The FSL method learns a classifier by using a limited number of training samples so as to identify different classes [26]. Meta-learning strategy is an effective learning strategy

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in FSL, which trains the model by a method called episode learning. Episode learning decomposes the dataset into different meta-tasks to complete the training of the model [27]. Because the FSL method based on the meta-learning strategy is very suitable for the situation where the samples in the dataset are limited, it can make the training of the model without the restriction of the number of labeled samples, so it has been successfully tried and applied in various fields, such as natural language processing, medical application, and computer vision [28], [29], [30].

In the field of HSI processing, Liu et al. [31] proposed a new method named deep FSL (DFSL) for solving the few-shot HSI classification problem, which used the FSL method based on the meta-learning strategy to make the network learn an embedding metric space, and finally completed the classification task in such a space. Gao et al. [32] proposed a deep classification model based on a relational network and used the meta-learning strategy as the training method. This method enhanced the learning ability of the network under small-sample conditions through meta-tasks generated from different datasets. Bai et al. [33] proposed a new few-shot classification model based on adaptive subspaces and feature transformation, which was trained through the meta-learning strategy, and finally classified the hyperspectral samples using a distance metric by an adaptive subspace classifier.

As an effective method to alleviate the insufficiency of labeled data, FSL based on the meta-learning strategy has received great attention in the field of HSI classification in recent years, but its application in HSI-CD has not been found yet. Therefore, how to make the FSL method based on the meta-learning strategy maximize its effectiveness in the HSI-CD task is a highly anticipated work.

B. Domain Adaptation

For most methods, the model can achieve high accuracy when given enough labeled samples as the training set. However, if the number of samples available to train is very limited, the performance of existing methods degrades dramatically. In order to cope with the situation where the HSI dataset to be processed (called target domain) cannot provide a large number of samples, it is feasible to supplement trainable data for the target domain through the learnable knowledge from additional datasets (called source domain), which is referred to as cross-domain HSI processing task.

Nevertheless, there is usually a problem of distribution inconsistency between the target domain and the source domain, which is often caused by changes in the shooting environment and differences in equipment. The distribution discrepancy of cross-domain data, that is, the domain shift problem will hinder the transfer and sharing of learnable information in the source domain, making it difficult for the model trained on the source domain to perform equally well on the target domain. In order to better solve the domain shift problem and make the model achieve the generalization from the source domain to the target domain, active research has been done and many methods have been proposed, including covariance-based methods [34], transfer learning fine-tuning-based methods [35], and domain adaptation-based methods, especially the domain adaptation methods have received more attention. Zhang et al. [36] focused on both the statistical features and the topological structure information of the data, and proposed a domain adaptation method that built a graph based on features and introduced graph optimal transport to align distributions, which effectively reduced the domain shift and improved the performance of the model for cross-scene HSI classification. Wang et al. [37] designed a model that could learn manifold embedding to achieve distribution matching between source and target domains, and the domain adaptation was accomplished while maintaining the manifold of the target domain without distortion in the matching process. A domain adaptation method based on neural embedding matching (NEM) was put forward by Wang et al. [38], which fit an intermediate representation space suitable for both the source and target domains by the NEM and matched the class distribution of both domains. Li et al. [39] designed a deep approach to implement domain adaptation, which gradually minimizes the distribution deviation between different domains through the maximum mean discrepancy criterion and the loss function in two stages, and finally built a common embedding space to improve model performance.

The above methods show that under the cross-scene condition, domain adaptation technology can make the network more discriminative and better generalizable. This article focuses on the excellent property that generative adversarial networks (GANs) can learn complex data distributions without relying on any prior assumptions [40]. At the same time, considering the scarcity of HSI-labeled samples, a BiGDA method is proposed to solve the domain shift problem between heterogeneous domains, which combines the GAN structure with the concept of FSL to minimize the discrepancy in the interdomain distribution between the two domains under the condition of few-shot to achieve domain adaptation.

III. Proposed Method

The overall framework of the BiG-FSLF and the detailed design of each part are explained in this section. The proposed BiG-FSLF mainly consists of feature extraction, CHFSL, and BiGDA. The simple procedure of the overall training and testing of BiG-FSLF is shown in Fig. 1.

We denote bitemporal VHRIs of the source domain as $\mathbf{X}_1^s, \mathbf{X}_2^s \in \mathbb{R}^{H\times W\times \mathrm{CH}}$, where H and W represent the altitude and breadth of the spatial domain of VHRIs, respectively. The CH represents the size of the spectral domain of VHRIs. We denote bitemporal HSIs of the target domain as $\mathbf{X}_1^t, \mathbf{X}_2^t \in \mathbb{R}^{h\times w\times \mathrm{ch}}$, where h and w represent the altitude and breadth of the spatial domain of HSIs. The ch represents the size of the spectral domain of HSIs. First, extract paired image blocks from $\mathbf{X}_1^s, \mathbf{X}_2^s$ of source domain and subtract them to obtain the difference image block $x_s \in \mathbb{R}^{p\times q\times \mathrm{CH}}$, where p and q are the height and width of the block. $\{x_s^m\}_{m=1}^N$ constitutes the source domain training dataset D_S, and N denotes the number of image blocks, and x_s^m denotes the mth block. The same operation is performed on the bitemporal images of the target domain to obtain the difference image block $x_t \in \mathbb{R}^{p\times q\times \mathrm{ch}}$, and $\{x_t^m\}_{m=1}^N$ constitutes the target domain dataset D_T, where

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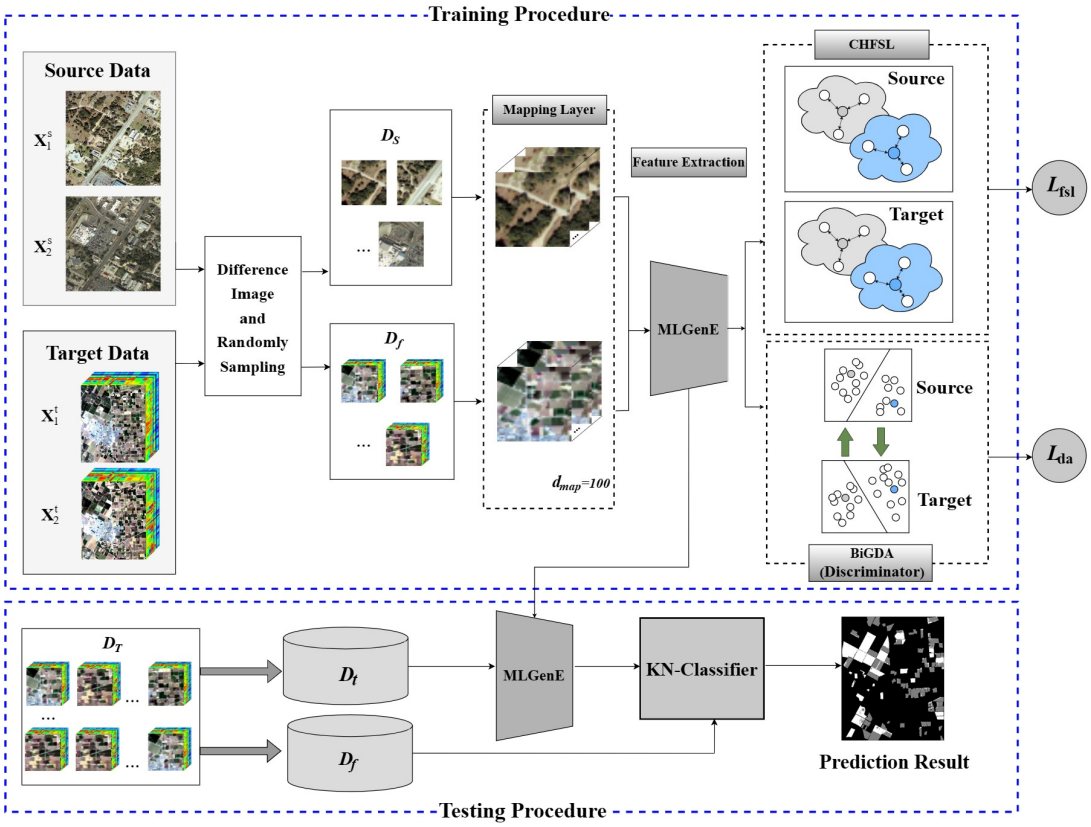


Fig. 1. Procedure of the BiG-FSLF.

x_t^m denotes the mth block. D_T is further divided into a few-shot dataset D_f for training and a test dataset D_t.

A. Overall Process of BiG-FSLF

In the training process, the data in D_S and D_f are processed as follows.

1) To unify the dimensions of the data from different sources, the x_s^m and the x_t^m are first sent to the mapping layer, which maps them to the same dimension. The mapping layer is a plain 2-D convolutional layer, which has a convolutional kernel size of 9×9. Thus, the original spectral dimensions of the source and target domains (d_s and d_t) can be unified to the d_{map} dimension (d_{map} is set to 100 in the experiment).
2) MLGenE extracts features from x_s^m and x_t^m after unified dimensions, and maps these features into the same metric space to obtain joint spatial–spectral embedded features of source domain data and target domain data (see Section III-B for details).
3) In the CHFSL stage, distance metrics are applied for the source domain embedded features and target domain embedded features, respectively, so that MLGenE learns the potential relationships between samples in the metric space (see Section III-C for details).
4) Meanwhile, the feature distributions of the data in source and target domains are reconstructed by MLGenE and subsequently discriminated by the discriminator, which completes domain adaptation at the feature distribution level through BiGDA (see Section III-D for details).

After training, MLGenE can produce feature distributions with high intraclass relevance and low interclass relevance, so in the test phase, the k-neighbors (KN) classifier fit by a few amount of data in the D_f can discriminate the changed and unchanged information of the data to be detected and finally complete the CD.

B. MLGenE Module

Since image blocks from different domains have different dimensions, in order to unify the dimensions, the mapping layer maps the original spectral images from the source domain and the target domain to a unified feature dimension d_{map}. Subsequently, MLGenE further processes the data. As shown in Fig. 2, a residual network and an encoder are connected to form the MLGenE.

The residual network is composed of two residual blocks, namely residual blocks 1 and 2. One residual block consists of three 3-D convolutional layers to extract features, and the extracted features are concatenated together via the skip connection and fed to the next residual block for the same operation. Finally, the features are mapped into a metric space for subsequent CHFSL (see Section III-C). The encoder in MLGenE further processes the features extracted by the residual network and then cooperates with the discriminator for subsequent BiGDA (see Section III-D). The encoder comprises two convolutional blocks. The first convolutional block

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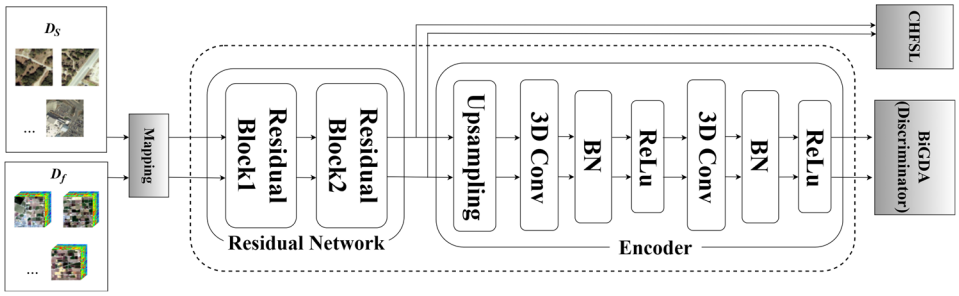


Fig. 2. Structure flowchart of MLGenE.

consists of an upsampling layer, a 3-D convolutional layer, a batch normalization (BN) layer, and a ReLU layer, and the second consists of a 3-D convolutional layer, a BN layer, and a ReLU layer.

C. CHFSL in BiG-FSLF

CHFSL is performed on the D_S and the D_f, where D_f consists of a very small number of labeled samples (e.g., five or ten samples). In order to avoid overfitting due to too few samples, a scheme of adding random Gaussian noise is adopted to expand the number of training samples in D_f.

In the training phase, FSL is performed alternately between the source and target domains, completing CHFSL. In order to enable MLGenE to learn more appropriate embedded features for the two domains, a form of meta-learning is used [41], that is, the few-shot task is simulated by minibatch sampling called episode, and in the iteration process, the network is updated by learning a series of episodes to learn embedded feature and make the model generalizable [42]. More specifically, CHFSL is achieved by alternately executing the training tasks of the source episodic and target episodic. A small number of samples are randomly selected from D_S to form a source episodic to simulate a few-shot task. Similarly, a small number of samples from D_f is selected to form a target episodic to simulate a few-shot task. Actually, the training iterations are carried out continuously through these few-shot tasks simulated by the cross-heterogeneous domain episodic. Take the example of a few-shot task simulated by the source episodic, we transform the CD task into a binary classification task that distinguishes the changed class and unchanged class. First, C classes are randomly selected from D_S, and then extract K labeled samples from each of these C classes. The extracted $C \times K$ samples form a support set, which is called the K-shot C-way task. So, the support set can be denoted as $S = \{(x_s^m, y_s^m)\}(C \times K/m = 1)$, where x_s^m is the mth input sample. At the same time, select N labeled samples from the C classes selected by the support set to form the query set. So the query set is denoted as $Q = \{(x_s^m, y_s^m)\}$ $(C \times N/m = 1)$. It is worth noting that the C classes that make up the support set and the query set are the same, but the samples of each class in the two sets are not the same, that is, $(S, Q \in D_S) \cup (S \cap Q = \emptyset)$. Such sampling forms a support set and a query set, which constitutes an episode for one FSL. Fig. 3 shows the source domain FSL process (one-shot two-way as an example). The features are extracted by MLGenE to obtain support features and query features,

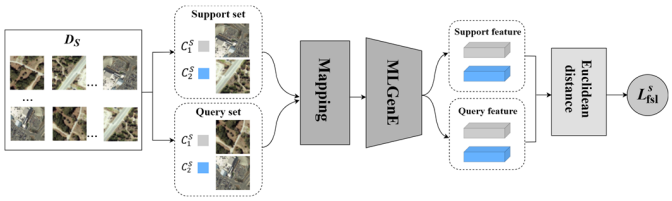


Fig. 3. Source domain FSL process. One-shot two-way as an example.

and the features are mapped to a metric space. By calculating the distance between features in the metric space, the class distribution of query samples is obtained, which is denoted as

$$p_\varphi(y = k|x) = \frac{\exp\left(-d\left(G_\varphi(x), c_k\right)\right)}{\sum_{k=1}^{C} \exp\left(-d\left(G_\varphi(x), c_k\right)\right)} \tag{1}$$

where c_k denotes the feature of the kth class in the support set, x denotes the feature of the query set sample, y is the label of x, C is the number of classes in each episode, $d(\cdot, \cdot)$ denotes the Euclidean distance function, and G_φ denotes the MLGenE with parameter φ.

Eventually, the loss function is obtained by taking the minimization of the negative of the class distribution function of the query samples, that is, the loss of the source domain FSL can be expressed as the negative log-probability of the true class k

$$L_{\text{fsl}}^s = E_{S_s, Q_s}\left(-\sum_{(x,y)\in Q_s} \log p_\varphi(y = k|x)\right) \tag{2}$$

where S_s is the support set composed of samples in D_S and Q_s is the query set from D_S.

The target domain episodic training task is the same as the source domain episodic training task. Select samples from D_f to compose the support and query sets, constituting one episode to perform one target domain FSL. The loss of all query set samples in the target domain FSL can be expressed as

$$L_{\text{fsl}}^t = E_{S_t, Q_t}\left(-\sum_{(x,y)\in Q_t} \log p_\varphi(y = k|x)\right) \tag{3}$$

where S_t is the support set composed of samples in D_f and Q_t is the query set composed of samples in D_f.

CHFSL enables MLGenE to learn both the shareable knowledge and personality feature of the source and target domains, so that the MLGenE can generate embedded features with high

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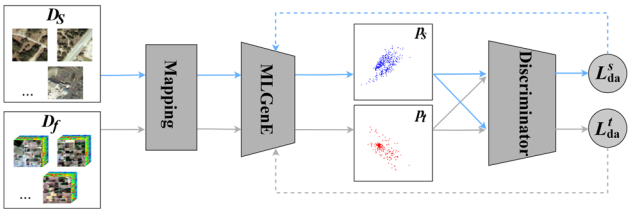


Fig. 4. BiGDA in BiG-FSLF.

intraclass relevance and low interclass relevance in the metric space after training.

D. BiGDA in BiG-FSLF

In order to solve the problem of hindering CHFSL due to the domain shift, a BiGDA approach is proposed to gradually reduce the difference of feature distribution between the source and target domains in a bidirectional generation way through the generative adversarial strategy to achieve domain adaptation. A simple procedure of BiGDA is given in Fig. 4, that is, the feature distributions of the source and target domains can be reconstructed by MLGenE to be more and more similar through the continuous game between MLGenE and the discriminator. Specifically, in one iteration of the training process, MLGenE first extracts the source and target domain data features and reconstructs the feature distributions to obtain the reconstructed distribution p_S of the source domain and the reconstructed distribution p_t of the target domain, and then the discriminator discriminates the reconstructed distributions, that is, $D(p_S, p_t)$, where D denotes the discriminator network, which consists of multiple convolutional layers, BN, and LeakyReLU, and finally connects a fully connected layer and outputs the discriminant result by softmax. In this iteration, the feature distribution of the source domain data is used as the benchmark, so that the feature distribution of the target domain data is closer to the source domain. In the next iteration, with the feature distribution of the target domain data as the benchmark, MLGenE reconstructs the feature distribution so that the reconstructed feature distribution of the source domain data is as close as possible to that of the target domain data, and then the discriminator performs discrimination. In the process of such reconstruction and discrimination, MLGenE can capture and reconstruct the common feature distribution of the source and target domains, while the discriminator becomes more and more difficult to distinguish the distribution differences between the two domains. Finally, MLGenE can minimize the feature distribution differences of the cross-domain data to achieve domain adaptation. The bidirectional generation process enables MLGenE to quickly learn the more universal feature distribution for two domains while avoiding overfitting under the few-shot condition.

Where the domain adaptation loss of the source domain is formalized as

$$L^s_{\text{da}} = \frac{1}{N}\sum_{m=1}^{N}\left(D_\beta\left(G_\varphi\left(x_t^m\right)\right) - D_\beta\left(G_\varphi\left(x_s^m\right)\right)\right) \tag{4}$$

where D_β denotes the discriminator network with parameter β, N denotes the total number of samples, x_s^m denotes the mth input sample in the source domain, and x_t^m denotes the mth input sample in the target domain.

In the BiGDA process, MLGenE is expected to maximize the error rate of the discriminator, while the discriminator is expected to minimize its own error rate, and the two of them are antagonistic to each other and progress together in competition.

Therefore, (4) can be divided into two subproblems for optimization

$$\arg\max_{\beta}\frac{1}{N}\sum_{m=1}^{N}\left(D_\beta\left(G_\varphi\left(x_t^m\right)\right) - D_\beta\left(G_\varphi\left(x_s^m\right)\right)\right) \tag{5}$$

and

$$\arg\min_{\varphi}\frac{1}{N}\sum_{m=1}^{N}D_\beta\left(G_\varphi\left(x_s^m\right)\right). \tag{6}$$

Similarly, the domain adaptation loss of the target domain can be formulated as

$$L^t_{\text{da}} = \frac{1}{N}\sum_{m=1}^{N}\left(D_\beta\left(G_\varphi\left(x_s^m\right)\right) - D_\beta\left(G_\varphi\left(x_t^m\right)\right)\right). \tag{7}$$

Furthermore, (7) can be divided into (8) and (9) to solve

$$\arg\max_{\beta}\frac{1}{N}\sum_{m=1}^{N}\left(D_\beta\left(G_\varphi\left(x_s^m\right)\right) - D_\beta\left(G_\varphi\left(x_t^m\right)\right)\right) \tag{8}$$

and

$$\arg\min_{\varphi}\frac{1}{N}\sum_{m=1}^{N}D_\beta\left(G_\varphi\left(x_t^m\right)\right). \tag{9}$$

E. Total Loss Function

From (2) and (4), the source domain total loss function is as follows:

$$L^s_{\text{total}} = \arg\ \min\max\ \ \lambda L^s_{\text{da}} + L^s_{\text{fsl}} \tag{10}$$

where λ is the regularization parameter used to weigh the loss term. The value of λ will be discussed in Section V-B.

Similarly, from (3) and (7), the target domain total loss function is as follows:

$$L^t_{\text{total}} = \arg\ \min\max\ \ \lambda L^t_{\text{da}} + L^t_{\text{fsl}}. \tag{11}$$

The final objective function L_{total} of the proposed BiG-FSLF model is

$$L_{\text{total}} = L^s_{\text{total}} + L^t_{\text{total}}. \tag{12}$$

Finally, the training flowchart of BiG-FSLF can be summarized as Algorithm 1.

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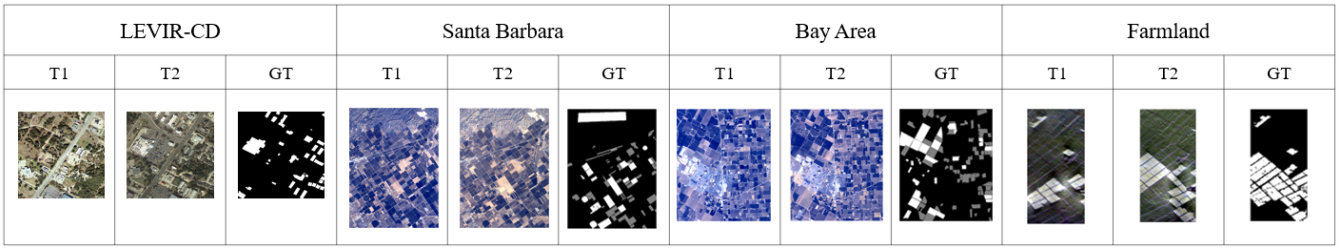


Fig. 5. Pseudo-color images of the experimental datasets and their ground-truth images.

Algorithm 1 Training Flowchart of BiG-FSLF

Input:
Source domain VHR difference image patches: $\{x_s^m\}_{m=1}^N$ and label $\{Y_s^m\}_{m=1}^N$;
Target domain HSI difference image patches: $\{x_t^m\}_{m=1}^N$ and label $\{Y_t^m\}_{m=1}^N$;
learning rate lr, training epochs E, Adam optimizer;
for each $epoch \in [1, E]$ **do**
if $epoch\%2 == 0$ **then**
1: Extract feature:
After Residual Network shown in Fig. 2,
1. 1: Gain source domain support set feature group $\{F_{ss}^l\}^m$ form x_s^m: $\{F_{ss}^l\}^m = \{\{F_{ss}^1, F_{ss}^2\}\}^m$;
1. 2: Gain source domain query set feature group $\{F_{sq}^l\}^m$ form x_s^m: $\{F_{sq}^l\}^m = \{\{F_{sq}^1, F_{sq}^2\}\}^m$;
1. 3: Gain target domain feature group $\{F_t^l\}^m$ form x_t^m: $\{F_t^l\}^m = \{\{F_t^1, \ldots, F_t^4\}\}^m$;
2: Source FSL:
Calculate the loss L_{fsl}^s by $\{F_{ss}^l\}^m, \{F_{sq}^l\}^m$ and $\{Y_s^m\}_{m=1}^N$, using the formulas given in (1) and (2);
3: After Encoder shown in Fig. 2,
3. 1: Gain reconstructed source domain feature distribution p_S from $\{F_{ss}^l\}^m$ and $\{F_{sq}^l\}^m$;
3. 2: Gain reconstructed target domain feature distribution p_t from $\{F_t^l\}^m$;
4: After Discriminator shown in Fig. 4,
Calculate the loss L_{da}^s, using the formulas given in (5) and (6);
5: Back-propagate $L_{\text{fsl}}^s + L_{\text{da}}^s$, and solve it with Adam;
else
1: Extract feature:
After Residual Network shown in Fig. 2,
1. 1: Gain target domain support set feature group $\{F_{ts}^l\}^m$ form x_t^m: $\{F_{ts}^l\}^m = \{\{F_{ts}^1, F_{ts}^2\}\}^m$;
1. 2: Gain target domain query set feature group $\{F_{tq}^l\}^m$ form x_t^m: $\{F_{tq}^l\}^m = \{\{F_{tq}^1, F_{tq}^2\}\}^m$;
1. 3: Gain source domain feature group $\{F_s^l\}^m$ form x_s^m: $\{F_s^l\}^m = \{\{F_s^1, \ldots, F_s^4\}\}^m$;
2: Target FSL:
Calculate the loss L_{fsl}^t by $\{F_{ts}^l\}^m, \{F_{tq}^l\}^m$ and $\{Y_t^m\}_{m=1}^N$, using the formulas given in (1) and (3);
3: After Encoder shown in Fig. 2,
3. 1: Gain reconstructed target domain feature distribution p_t from $\{F_{ts}^l\}^m$ and $\{F_{tq}^l\}^m$;
3. 2: Gain reconstructed target domain feature distribution p_S from $\{F_s^l\}^m$;
4: After Discriminator shown in Fig. 4,
Calculate the loss L_{da}^t, using the formulas given in (8) and (9);
5: Back-propagate $L_{\text{fsl}}^t + L_{\text{da}}^t$, and solve it with Adam;
end if
end for

IV. Experiments and Analysis

In this article, three commonly used evaluation indices, including overall accuracy (OA), average accuracy (AA), and Kappa coefficient, are used to evaluate the CD performance of different methods. Where OA denotes the true classification ratio of the whole pixel, AA denotes the AA of all categories, and the kappa coefficient reflects the agreement between the classification results and the ground-truth image, the higher its value indicates the better performance of CD. The OA can be calculated by the following equation:

$$\text{OA} = \frac{\text{TP} + \text{TN}}{\text{TP} + \text{TN} + \text{FP} + \text{FN}} \tag{13}$$

where TP, TN, FP, and FN denote the number of correctly predicted changed pixel points and the unchanged pixel points, the number of wrongly predicted unchanged pixel points, and the changed pixel points, respectively.

The AA calculation formula is as follows:

$$\text{AA} = \frac{\sum_{i=1}^{n} \text{accuracy}_i}{n} \tag{14}$$

where n denotes the number of classifications and accuracy_i is the classification accuracy of the ith class.

The Kappa coefficient can be calculated by the following equations:

$$\begin{cases} P_{\text{TP}} = (\text{TP} + \text{FP}) \times (\text{TP} + \text{FN}) \\ P_{\text{TN}} = (\text{TN} + \text{FP}) \times (\text{TN} + \text{FN}) \\ \text{PE} = \dfrac{P_{\text{TP}} + P_{\text{TN}}}{(\text{TP} + \text{TN} + \text{FP} + \text{FN})^2} \\ \text{Kappa} = \dfrac{\text{OA} - \text{PE}}{1 - \text{PE}}. \end{cases} \tag{15}$$

A. Experimental Datasets

To make an objective and fair evaluation of the BiG-FSLF, three challenging HSI datasets are selected for experiments, and their performance is analyzed and discussed. One image in the LEVIR-CD [43] dataset is used as the source domain, and three HSI datasets Santa Barbara [44], Bay Area [44], and Farmland [45] are used as the target domain. For detailed information, see Table I. The pseudo-color images and ground-truth images are given in Fig. 5.

B. Compared Methods and Experimental Setting

To verify the effectiveness of the proposed method, we compare our proposed method with some recently proposed methods, including support vector machine (SVM) [46], fast dense spectral–spatial convolution network (FDSSC) [47], double-branch multi-attention mechanism network (DBMA) [48], double-branch dual-attention mechanism network (DBDA) [49], DFSL [31], deep cross-domain few-shot learning network (DCFSL) [50], spectral-spatial siamese network (S3Net) [51], multilevel encoder-decoder attention network (ML-EDAN) [52], and cross-temporal interaction symmetric attention network (CSANET) [53]. In the experiment, we select the same number of labeled samples for training to ensure fairness,

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TABLE I
DATASET INFORMATION

Dataset	LEVIR-CD	Santa Barbara	Bay Area	Farmland
Size	1024×1024	980×740	600×500	450×140
Spectral Band	3	224	224	155
Data T1	2002	2013	2013	2006
Data T2	2013	2014	2015	2007
Sensor	Google Earth	AVIRIS	AVIRIS	EO-1
Type	VHRI	HSI	HSI	HSI

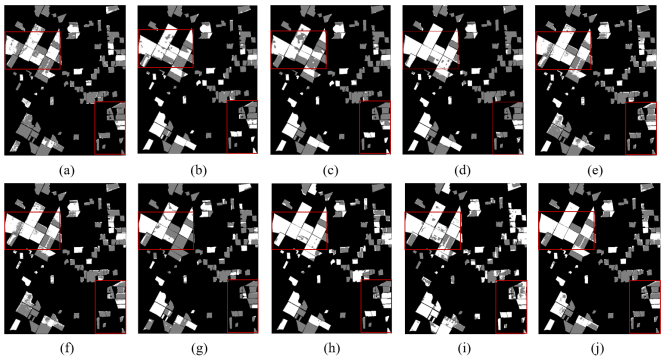


Fig. 6. Final change maps obtained using different methods for the Bay Area dataset. (a) SVM. (b) FDSSC. (c) DBMA. (d) DBDA. (e) DFSL. (f) DCFSL. (g) S3Net. (h) ML-EDAN. (i) CSANET. (j) BiG-FSLF.

so the experiment is set to take five labeled samples from each of the changed and unchanged classes. To adapt to the cross-domain situations, 500 labeled samples from each class in the source domain dataset are randomly selected to provide transferable knowledge for target domain detection.

In BiG-FSLF, the input patch size is set to 9 × 9, the learning rate is set to 0.001, and the number of training iterations is set to 2000. The learnable parameter of the activation function is empirically set to 0.2. For the FSL process, we set one labeled sample per class in the support set and 19 labeled samples per class in the query set of each episode. Our proposed BiG-FSLF is implemented on the PyTorch framework, and the experiments are performed using an NVIDIA GeForce RTX 3080 Ti personal computer.

C. Experimental Results and Analysis

1) Results on Bay Area Dataset and Santa Barbara Dataset: Since the two datasets, the Bay Area dataset and the Santa Barbara dataset, are captured by the same sensor and cover similar scenes, these two datasets will be discussed together. Fig. 6 shows the experimental results on the Bay Area dataset, and Fig. 7 shows the results on the Santa Barbara dataset, with quantitative evaluations in Tables II and III, respectively. Among them, the results of the SVM approach on the two datasets are not very different, and the achieved accuracy ranks moderately among all the compared methods, proving that the SVM is suitable for the few-shot condition, but the achieved accuracy is still difficult to surpass some deep learning methods. For FDSSC, DBDA, DBMA, ML-EDAN, and CSANET, which are deep learning methods, FDSSC, DBDA, and DBMA perform better in both datasets, mainly because the boundaries of these two datasets are more regular, and the regions of changed and unchanged are more uniform.

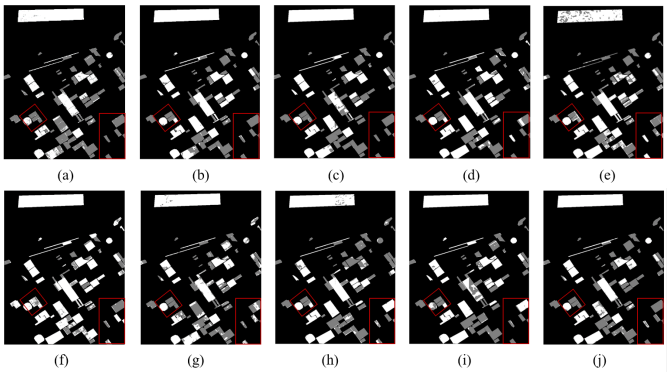


Fig. 7. Final change maps obtained using different methods for the Santa Barbara dataset. (a) SVM. (b) FDSSC. (c) DBMA. (d) DBDA. (e) DFSL. (f) DCFSL. (g) S3Net. (h) ML-EDAN. (i) CSANET. (j) BiG-FSLF.

TABLE II
SPECIFIC EVALUATION INDEX VALUES OBTAINED BY VARIOUS METHODS ON THE BAY AREA DATASET

Method	OA	AA	Kappa
SVM	0.8584	0.8690	0.7163
FDSSC	0.8899	0.8709	0.7305
DBMA	0.8583	0.8350	0.6664
DBDA	0.8854	0.8674	0.7238
DFSL	0.8420	0.8345	0.6783
DCFSL	0.8665	0.8604	0.7287
S3Net	0.6562	0.6598	0.3179
ML-EDAN	0.7804	0.7199	0.4346
CSANET	0.7825	0.7692	0.5295
BiG-FSLF	**0.9376** (↑4.77%)	**0.9383** (↑6.74%)	**0.8748** (↑14.43%)

However, the performance of ML-EDAN in Santa Barbara is quite different from that in Bay Area, which indicates that the method is less robust to different scenarios. The superiority of CSANET performance depends more on the selection of the number of training samples. In the case of using very few samples, the effect of the model is relatively general. For S3Net, DFSL, and DCFSL, which are methods aiming at solving the few-shot problem, S3Net is affected by the number of PCA band selections and the setting of the patch size, and the differences in the results achieved on different datasets are great. Compared with S3Net, DFSL, and DCFSL have achieved better results, confirming that the FSL method based on the meta-learning strategy is more helpful to solve few-shot problems. Our proposed method achieves the best results on both datasets, proving the effectiveness of our method for few-shot HSI-CD. Compared with the DCFSL method which is also cross-domain, BiG-FSLF has better results. The reason for the performance gap may be that our method achieves better domain adaptation, which can be more conducive to the transfer of source domain data knowledge and can better implementation of changed region detection. It can be seen from Tables II and III that compared with the optimal results in the comparison methods, our method improves OA by 4.77% and 1.87% and Kappa by 14.43% and 5.69%, respectively. For both datasets, BiG-FSLF can get similar results, which proves the stability and superiority of BiG-FSLF.

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TABLE III
SPECIFIC EVALUATION INDEX VALUES OBTAINED BY VARIOUS METHODS ON THE SANTA BARBARA DATASET

Method	OA	AA	Kappa
SVM	0.8568	0.8574	0.7134
FDSSC	0.9049	0.8842	0.7664
DBMA	0.9002	0.8799	0.7548
DBDA	0.9138	0.9040	0.8015
DFSL	0.8257	0.8240	0.6501
DCFSL	0.8622	0.8327	0.6973
S3Net	0.8234	0.8103	0.6241
ML-EDAN	0.8393	0.8265	0.6488
CSANET	0.8502	0.8312	0.6395
BiG-FSLF	**0.9325** (↑1.87%)	**0.9290** (↑2.5%)	**0.8584** (↑5.69%)

TABLE IV
SPECIFIC EVALUATION INDEX VALUES OBTAINED BY VARIOUS METHODS ON THE FARMLAND DATASET

Method	OA	AA	Kappa
SVM	0.8836	0.8564	0.7468
FDSSC	0.9172	0.9012	0.8062
DBMA	0.9157	0.8957	0.7986
DBDA	0.9084	0.8909	0.7796
DFSL	0.9042	0.8667	0.7755
DCFSL	0.9248	0.8993	0.8139
S3Net	0.8655	0.8298	0.6560
ML-EDAN	0.8003	0.7584	0.5311
CSANET	0.9141	0.8900	0.7962
BiG-FSLF	**0.9375** (↑1.27%)	**0.9284** (↑2.72%)	**0.8496** (↑3.57%)

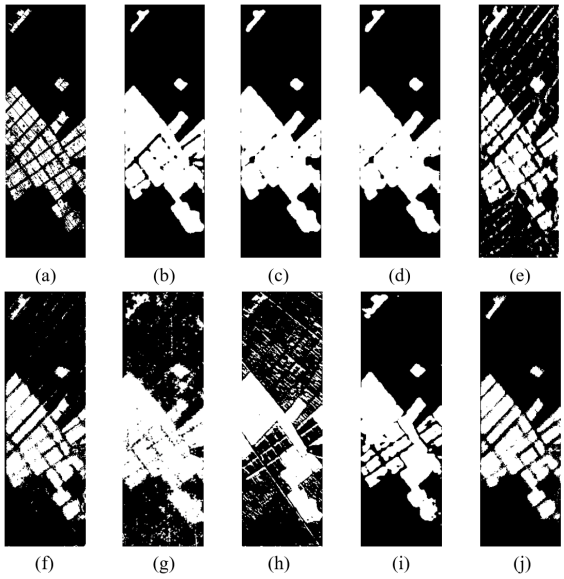


Fig. 8. Final change maps obtained using different methods for the Farmland dataset. (a) SVM. (b) FDSSC. (c) DBMA. (d) DBDA. (e) DFSL. (f) DCFSL. (g) S3Net. (h) ML-EDAN. (i) CSANET. (j) BiG-FSLF.

2) Results on Farmland Dataset: The outcomes of the nine comparison methods and the BiG-FSLF method on the Farmland dataset are shown in Fig. 8, and the values of the specific evaluation indicators obtained by the different methods are shown in Table IV.

It can be seen from the visualization that SVM is poor for the detection of changed regions, and there are many white noise points and many false detection regions in the visualizations of DFSL, ML-EDAN, and S3Net. The performance of FDSSC, DBDA, DBMA, and CSANET is not much different, although the generated result maps are less noisy and cleaner, but the detection of changed regions and unchanged regions is rougher, and there are still many false detections of unchanged regions. DCFSL achieves the superior result, but compared with our proposed method, the edge of the change regions generated by BiG-FSLF is relatively flat, with less noise, and the CD result is more accurate. In contrast, our proposed method achieves state-of-the-art performance with the support of CHFSL and BiGDA. As shown in Table IV, the OA of the methods, including ML-EDAN and S3Net, is below 90%, indicating that the CD accuracy of these methods is low. Among the methods that achieve better results, the OA of DCFSL reaches 92.48% and Kappa reaches 81.39%, indicating that FSL has a positive impact on CD accuracy with very few training samples. The proposed BiG-FSLF outperforms all the comparison methods on the Farmland dataset, and compared with the DCFSL, OA improves by 1.27% and Kappa improves by 3.57%, which proves its effectiveness.

D. Ablation Analysis

The main components of the proposed framework are CHFSL and BiGDA. To evaluate the contribution of the key components of BiG-FSLF, ablation analysis is performed by removing each component from the whole framework. There are three types of ablation models: 1) "BiG-FSLF (Baseline)": remove CHFSL, BiGDA, and source domain data from BiG-FSLF; 2) "BiG-FSLF (no-CHFSL)": remove CHFSL from BiG-FSLF; and 3) "BiG-FSLF (no-BiGDA)": remove BiGDA from BiG-FSLF. We conduct ablation experiments on three datasets for each of the ablation models, and the experimental results for these three models correspond to columns a–c. Obviously, the overall BiG-FSLF is superior to other ablation models and has been greatly improved. The "BiG-FSLF (Baseline)" model shows the worst results, i.e., good detection results could not be achieved only through the mapping layer and MLGenE. Furthermore, from the experimental results of the "BiG-FSLF (no-CHFSL)" ablation model given in column b of Table V, it can be seen that without CHFSL, the values of each evaluation index of BiG-FSLF are significantly degraded, confirming that CHFSL can make the model learn better under the condition of few-shot, and it is helpful for the model to learn the transferable knowledge of source domain and capture the personality features of the target domain. From the experimental results of the "BiG-FSLF (no-BiGDA)" ablation model given in column c of Table V, it can be seen that it is difficult to achieve the expected performance on the target domain without BiGDA. BiGDA can help heterogeneous domain data to align the feature distribution, which has a positive contribution to the CD of BiG-FSLF. Especially

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TABLE V
ABLATION COMPARISON OF EACH VARIANT OF BIG-FSLF (FIVE LABELED SAMPLES FROM TARGET DOMAIN)

	(a)	(b)	(c)	(d)	(e)
Model	BiG-FSLF (Baseline)	BiG-FSLF (no-CHFSL)	BiG-FSLF (no-BiGDA)	BiG-FSLF (no-Source Domain)	BiG-FSLF
Dataset	Bay Area				
OA(%)	76.31	92.77	91.49	90.06	**93.76**
Kappa	42.70	85.40	82.96	79.99	**87.48**
Dataset	Santa Barbara				
OA(%)	77.73	91.51	91.15	85.18	**93.25**
Kappa(%)	50.10	73.50	81.13	57.87	**85.84**
Dataset	Farmland				
OA(%)	81.75	92.78	92.89	91.19	**93.75**
Kappa(%)	46.15	83.42	82.14	80.18	**84.96**

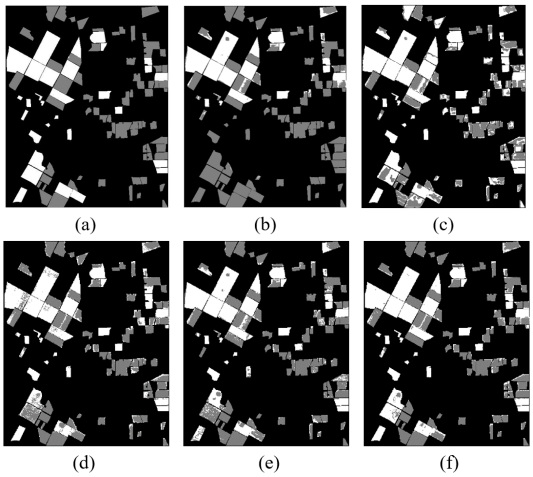


Fig. 9. Ablation analysis result images on the Bay Area dataset. (a) Ground truth map. (b) BiG-FSLF (baseline). (c) BiG-FSLF (no-CHFSL). (d) BiG-FSLF (no-BiGDA). (e) BiG-FSLF (no-source domain). (f) BiG-FSLF.

for Kappa, the index for measuring classification accuracy, there has been a significant improvement. Fig. 9(a) is the ground truth map of the Bay Area dataset. Fig. 9(b) shows the CD results of the model "BiG-FSLF (Baseline)" on the BayArea dataset. Fig. 9(c) gives the ablation results of model "BiG-FSLF (no-CHFSL)" on the Bay Area dataset, while Fig. 9(d) is the ablation results of model "BiG-FSLF (no-BiGDA)" on this dataset.

In order to confirm that the knowledge of the VHRI data of the source domain is instructive and helpful for CD of HSI of the target domain, we add a set of experiments to delete the source domain data, and the results are given in column d of Table V. "BiG-FSLF (no-source domain)" only uses the data of target domain for CD, and the ablation results show that the knowledge of the source domain data is indeed helpful and instructive for CD of the target domain, and also prove that the learnable information of the source domain data has achieved effective cross heterogeneous domain transfer generalization. Fig. 9(e) gives the results of BiG-FSLF when only the data of the target domain Bay Area dataset is available.

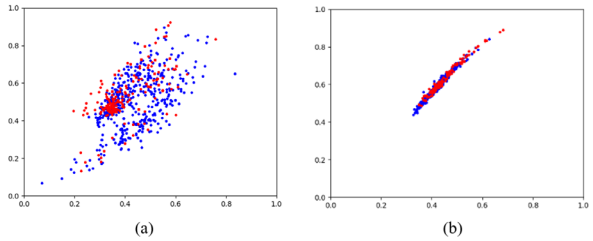


Fig. 10. Visual comparison of the Bay Area features before and after BiGDA. (a) Before. (b) After.

V. DISCUSSION

A. Domain Distribution Alignment Performance of BiGDA

The proposed BiGDA achieves feature distribution alignment of heterogeneous domain data, which lays the foundation for the utilization of source domain data and enables sharing of generalizable knowledge. To illustrate the effective implementation of domain adaptation, Fig. 10 takes the Bay Area dataset as an example and shows the visualization results of the distribution of feature vectors in the metric space. Fig. 10(a) shows the visualization result before the implementation of domain adaptation, while Fig. 10(b) shows the result after the implementation of domain adaptation. The blue scattered points are the feature distribution of the source domain data and the red scattered points are the feature distribution of the target domain data. It can be seen from the figure that before domain adaptation, the feature distributions of heterogeneous data have large differences, the feature distribution of the source domain is more dispersed, while the feature distribution of the target domain is mainly concentrated at the left center point. After the domain adaptation, the difference between the feature distributions of the two domains decreases and tends to be the same. The effectiveness of the proposed BiGDA method is verified, which can help mitigate the domain shift problem caused by cross-domain and facilitate knowledge transfer so that the source domain data can be used to maximum effect.

B. Hyperparameter Analysis

There is a regularization parameter λ in BiG-FSLF, and experiments are conducted to test its impact on the CD performance. The parameter values are selected as 10^{-4}, 10^{-3}, 10^{-2}, 10^{-1}, and 1. Fig. 11 shows the OA achieved by BiG-FSLF for the three datasets with different values of λ parameter settings. It can be seen from the figure that on the Bay Area dataset, BiG-FSLF can reach the highest OA when $\lambda = 0.01$, and then the OA decreases rapidly when $\lambda > 0.01$, so for the Bay Area dataset, λ is reasonably set to 0.01. Similarly, to achieve the best OA, $\lambda = 0.01$ and 0.001 are the best choices for the Santa Barbara dataset and the Farmland dataset, respectively. For the experiments on each dataset in this article, the λ value is set to the corresponding optimal value.

C. Discussion of the Number of Labeled Samples

We train the network with a different number of samples for analyzing its impact on the CD performance. Table VI shows the OA and Kappa obtained by different methods when trained with a different number of samples (three-to-seven per class) on the Farmland dataset. It is clear from Fig. 12(a) that

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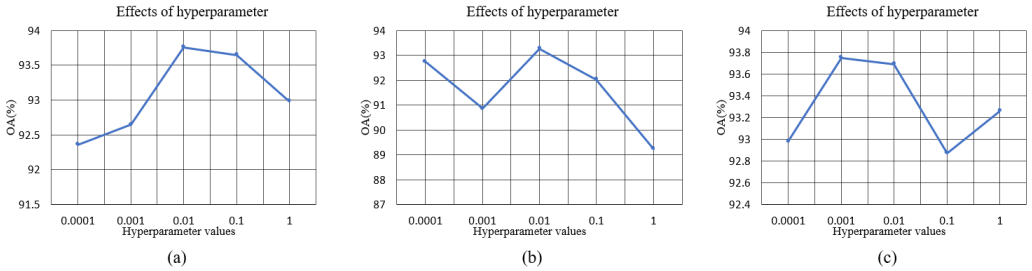


Fig. 11. Effects of λ on CD performance. (a) Bay Area. (b) Santa Barbara. (c) Farmland.

TABLE VI
DIFFERENT NUMBER OF TRAINING SAMPLES ON CD (THREE TO SEVEN LABELED SAMPLES PER CLASS FROM TARGET DOMAIN)

Method	SVM	FDSSC	DBMA	DBDA	DFSL	DCFSL	S3Net	ML-EDAN	CSANET	BiG-FSLF
number	**3**									
OA(%)	87.99	91.38	90.36	90.68	83.53	91.56	74.96	66.45	91.28	**93.07**
Kappa(%)	67.38	78.63	76.25	75.59	65.37	80.06	20.11	37.73	78.92	**83.32**
number	**4**									
OA(%)	90.84	91.41	91.54	90.81	89.13	91.51	83.25	68.55	85.69	**93.48**
Kappa(%)	75.57	78.60	76.25	75.59	75.62	80.08	57.70	43.69	60.59	**84.33**
number	**5**									
OA(%)	88.36	91.72	91.57	90.84	90.42	92.48	86.55	80.03	91.41	**93.75**
Kappa(%)	74.68	80.62	79.86	77.96	77.55	81.39	65.60	53.11	79.62	**84.96**
number	**6**									
OA(%)	93.08	92.62	92.13	91.92	92.12	92.36	93.48	80.56	92.01	**93.91**
Kappa(%)	84.43	81.57	80.61	79.95	80.97	81.56	84.58	60.09	81.76	**85.18**
number	**7**									
OA(%)	93.70	94.00	92.70	92.28	92.35	92.91	92.10	81.86	93.87	**94.04**
Kappa(%)	85.54	85.49	82.41	81.67	81.48	82.46	79.99	60.78	85.35	**85.91**

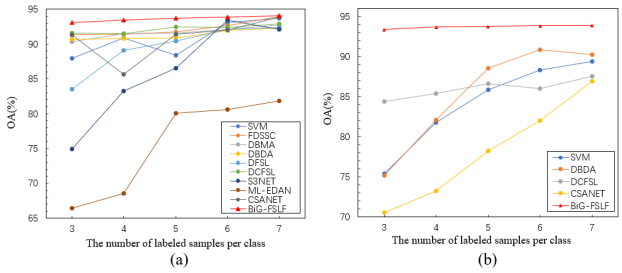


Fig. 12. Effects of a different number of training samples on CD performance. (a) Farmland. (b) Bay Area.

as the number of samples increase, the OA of most methods continue to improve, while our method has achieved optimal results for a different number of samples. In order to see more clearly, we selected several methods that performed well on the Farmland dataset and further compared them on the Bay Area dataset, as shown in Fig. 12(b). The results confirm that BiG-FSLF can adapt to the variation in the number of labeled samples and achieve good results under the few-shot condition.

VI. CONCLUSION

In this article, in order to address the problem of insufficient labeled training samples in the HSI-CD task, a CHFSL framework based on bidirectional generation is proposed, which attempts to assist HSI in CD by migrating VHRI data knowledge and alleviating the problem of domain shift caused by cross heterogeneous domain. The network mainly consists of a mapping layer for unifying dimensions, an MLGenE, and a discriminator for discriminating the feature distribution properties of heterogeneous domain data. This is the first time that the FSL mechanism combined with the generative adversarial strategy is introduced into the HSI-CD field. An MLGenE that can effectively capture interdomain common distribution features and intradomain embedded features of heterogeneous domain data is designed, which can maximize interclass differences while minimizing interdomain distribution differences. The proposed framework achieves the transfer of learnable information in VHRI and learns the discriminative features of HSI through the CHFSL. While the feature distributions of heterogeneous domain data are aligned by the BiGDA to make the VHRI data and HSI data have more similar embedding representations. Extensive comparison experiments and ablation experiments are conducted on three commonly used HSI-CD datasets, all of which show the effectiveness of the proposed BiG-FSLF in dealing with the few-shot problem and the domain shift problem, and it can show great detection performance with just five training samples.

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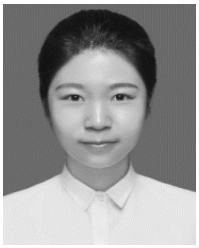
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